

Correlation coefficient (r)
and the
Coefficient of Determination (R^2)

MATH 213

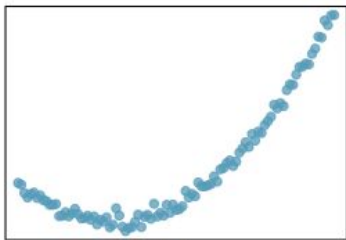
Measuring strength of association

- The *correlation* is a statistic that measures the strength of the *linear association* between two variables.
- It takes values between -1 (perfect negative) and +1 (perfect positive).
- A value of 0 indicates no linear association.

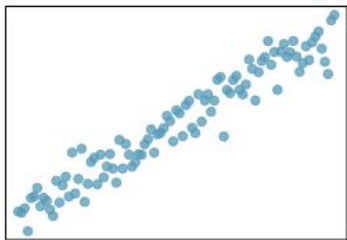
Play a few rounds of [Guess the Correlation](#) to develop some intuition for this statistic.

Assessing the correlation

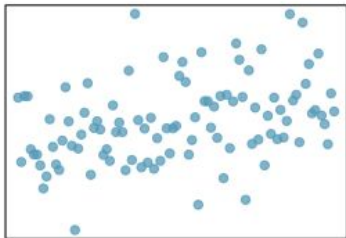
Which of the following has the strongest correlation, i.e. correlation coefficient closest to +1 or -1?



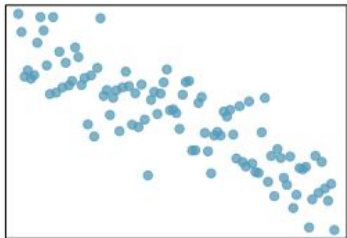
(a)



(b)



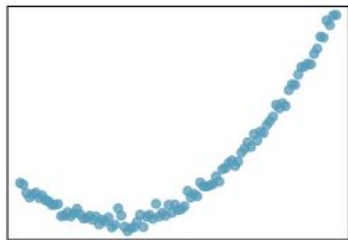
(c)



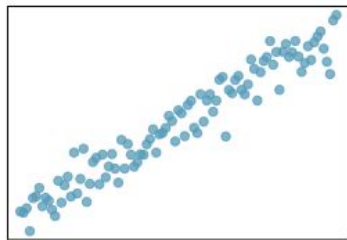
(d)

Assessing the correlation

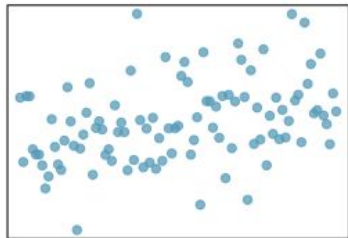
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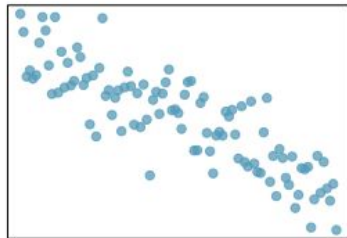
(a)



(b)



(c)

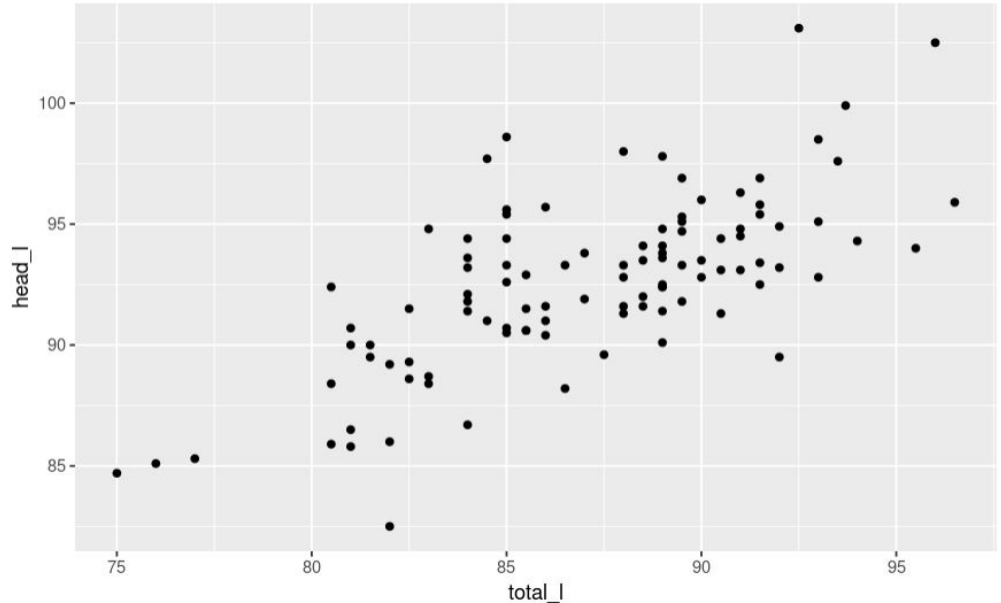


(d)

*(b) → correlation
means linear
association*

Possums

Guess the correlation coefficient for the relationship between head length and total length?



R^2 : The Coefficient of Determination

- The strength of the fit of a linear model is most commonly evaluated using R^2 .
- It tells us what percentage of the variability in the response variable is explained by the model.

Example: Calculate the R^2 value for the Possums Model

Variance of the head length variable
 $= (3.57)^2$

Variance of the residuals $= (2.58)^2$

R^2 = % reduction in variability of head length

$$= [(3.57)^2 - (2.58)^2] / (3.57)^2 = 0.477$$

About 48% of the variability in head length is explained by using total length as a predictor.

R^2 : The Coefficient of Determination

- So, about 48% of the variability in head length is explained by the model. The remainder of the variability is explained by variables not included in the model or by inherent randomness in the data.
- It turns out that R^2 is calculated as the square of the correlation coefficient.

$$r^2 = (0.69)^2 = 0.477 = R^2$$

Locating R^2 in Model Summary Output

```
Call:
lm(formula = head_1 ~ total_1, data = possum)

Residuals:
    Min       1Q   Median       3Q      Max
-7.1877 -1.5340 -0.3345  1.2788  7.3968

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  42.70979     5.17281   8.257 5.66e-13 ***
total_1       0.57290     0.05933   9.657 4.68e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.595 on 102 degrees of freedom
Multiple R-squared:  0.4776,    Adjusted R-squared:  0.4725
F-statistic: 93.26 on 1 and 102 DF,  p-value: 4.681e-16
```